# CSA3710 SOFTWARE TESTING
# EXPERIMENTS

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**QUESTIONS:**

**2. To write a case study to identify & analyze the requirements of ARMS Portals and to write a positive Test Scenarios as well as Test cases for the above application**

**Code:**

```
package kbexp;
public class kb {
   public static void main(String[] args) {

      String[] items = {
         "Home", "My Course", "Enrollment", "Attendance",
         "Assignment", "Examination", "Financial Record",
         "Disciplinary Record", "Raise Infra Issue", "My Profile"
      };

      String[] testCases = {
         "Check holiday notification",
         "View course status",
         "Select enrollment slot",
         "View attendance report",
         "View assignment details",
         "View examination details",
         "Check payment history",
         "View disciplinary details",
         "Submit infrastructure issue",
         "View profile details"
      };

      System.out.println("ARMS PORTAL - POSITIVE TEST CASES");
      System.out.println("-----------------------------------------------");

      System.out.printf("%-5s %-25s %-35s %-10s%n",
            "S.No", "Item", "Test Case", "Status");
```

```
        System.out.println("-----------------------------------------------");

        for (int i = 0; i < items.length; i++) {
            System.out.printf("%-5d %-25s %-35s %-10s%n",
                    i + 1, items[i], testCases[i], "PASS");
        }

        System.out.println("-----------------------------------------------");
        System.out.println("Test Approach: Manual Testing");
        System.out.println("Test Environment: Browser");
        System.out.println("Training Needed: No");
        System.out.println("Risk: Low");
        System.out.println("Test cases documented successfully.");
    }
}
```

**Report table:**

| S. No. | Item to be tested | Features | Approach | Test Data | Test Environment | Training Needed | Risk | Test Case / Scenario | Expected Output | Pass/Fail |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Home | Holiday notification | Manual | No | Laptop/Browser | No | Low | Open ARMS and check the home page | Home page and notifications appear | Pass |
| 2 | My Course | Course status | Manual | No | Laptop/Browser | No | Low | Open My Course | Course status is displayed | Pass |
| 3 | Enrollment | Enrollment slot | Manual | Valid slot | Laptop/Browser | No | Low | Select an available slot | Slot is selected successfully | Pass |
| 4 | Attendance | Attendance report | Manual | Student details | Laptop/Browser | No | Low | Open attendance report | Attendance details appear | Pass |
| 5 | Assignment | Assignment details | Manual | Valid course | Laptop/Browser | No | Low | Open an assignment | Assignment details appear | Pass |
| 6 | Examination | Examination details | Manual | Student details | Laptop/Browser | No | Low | View examination details | Examination information appears | Pass |
| 7 | Financial Record | Payment history | Manual | Student account | Laptop/Browser | No | Low | Open payment history | Payment details appear | Pass |
| 8 | Disciplinary Record | Disciplinary details | Manual | Student account | Laptop/Browser | No | Low | Open disciplinary record | Record details appear, if available | Pass |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| 9 | Raise Infra Issue | Submit issue | Manual | Valid issue details | Laptop/Browser | No | Low | Submit an infrastructure issue | Issue is submitted successfully | Pass |
| 10 | My Profile | Profile details | Manual | Valid student account | Laptop/Browser | No | Low | Open My Profile | Profile details appear | Pass |

**Result:** Positive test scenarios for the ARMS Portal were documented. Pass labels are illustrative and should be confirmed by testing the actual portal.

**Code** **output:**

SWlab.java | package-info.java | kb.java | package-info.java | package-info.java | kb.java

```
        System.out.println("ARMS PORTAL - POSITIVE TEST CASES");
        System.out.println("-------------------------------------------------");

        System.out.printf("%-5s %-25s %-35s %-10s%n",
                "S.No", "Item", "Test Case", "Status");

        System.out.println("-------------------------------------------------");

        for (int i = 0; i < items.length; i++) {
            System.out.printf("%-5d %-25s %-35s %-10s%n",
                    i + 1, items[i], testCases[i], "PASS");
        }

        System.out.println("-------------------------------------------------");
        System.out.println("Test Approach: Manual Testing");
        System.out.println("Test Environment: Browser");
        System.out.println("Training Needed: No");
        System.out.println("Risk: Low");
        System.out.println("Test cases documented successfully.");
    }
}
```

Problems | Servers | Terminal | Data Source Explorer | Properties | Console | Sponsor the Eclipse IDE

```
<terminated> kb [Java Application] C:\Users\SAIL\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.12.v20260826-1216\jre\bin\javaw.exe
3     Enrollment              Select enrollment slot          PASS
4     Attendance              View attendance report          PASS
5     Assignment              View assignment details         PASS
6     Examination             View examination details        PASS
7     Financial Record        Check payment history           PASS
8     Disciplinary Record     View disciplinary details       PASS
9     Raise Infra Issue       Submit infrastructure issue     PASS
10    My Profile              View profile details            PASS
-------------------------------------------------
Test Approach: Manual Testing
Test Environment: Browser
Training Needed: No
Risk: Low
Test cases documented successfully.
```

**3. To write a case study to identify & analyze the requirements of ARMS Portals and to write a negative Test Scenarios as well as Test cases for the above application.**

**Code:**

```
package kbexp;
public class ARMSNegativeTest {
   public static void main(String[] args) {

      String[] items = {
         "Home",
         "My Course",
         "Enrollment",
         "Attendance",
         "Assignment",
         "Examination",
         "Financial Record",
         "Disciplinary Record",
         "Raise Infra Issue",
         "My Profile"
      };

      String[] testCases = {
         "Missing holiday information",
         "Invalid course details",
         "Invalid enrollment slot",
         "Unavailable attendance data",
         "Invalid assignment",
         "Expired examination link",
         "Invalid payment details",
         "Unauthorized record access",
         "Empty issue details",
         "Invalid profile information"
      };

      System.out.println("ARMS PORTAL - NEGATIVE TEST CASES");

      for (int i = 0; i < items.length; i++) {
         System.out.println("\nTest Case " + (i + 1));
         System.out.println("Item: " + items[i]);
         System.out.println("Scenario: " + testCases[i]);
```

```
            System.out.println("Expected Result: Error handled correctly");
        }

        System.out.println("\nNegative test cases documented successfully.");
    }
}
```

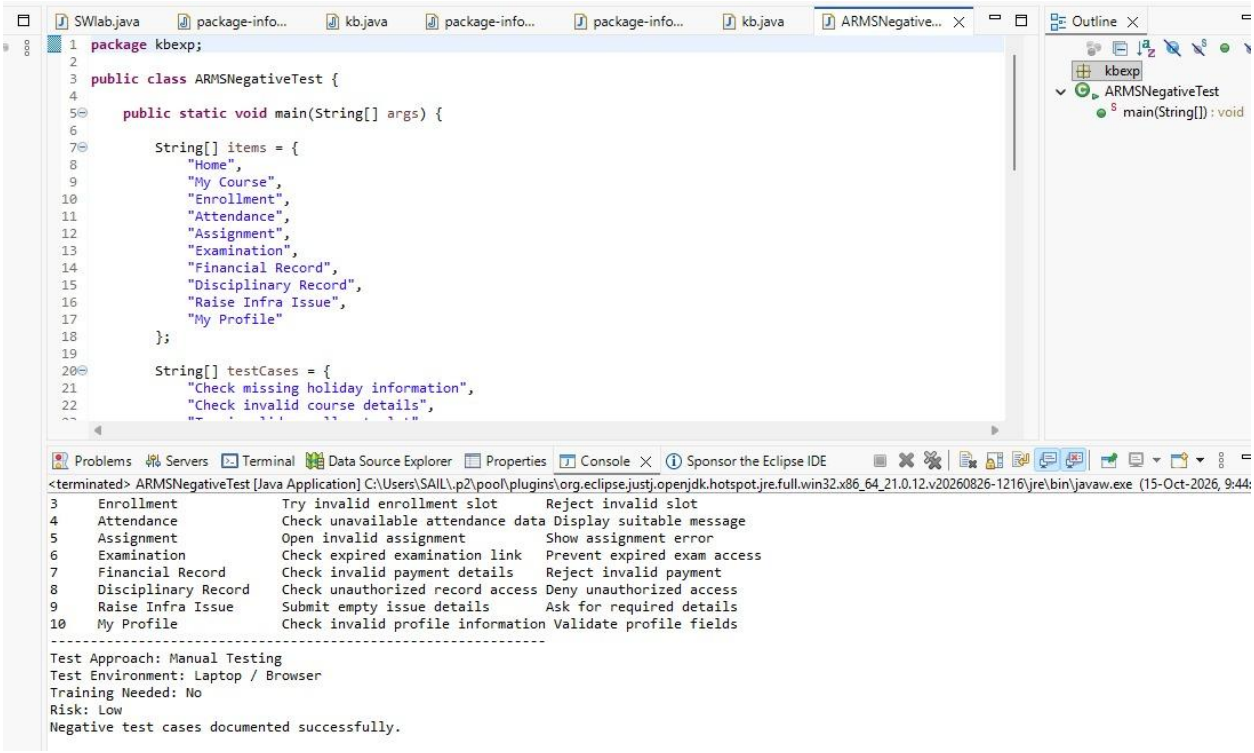
**Report:**

| S.No. | Item to be tested | Features | Approach | Test Data | Test Environment | Training Needed | Risk | Test Case / Scenario | Expected Output | Pass/Fail |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Home | Notifications | Manual | Missing data | Laptop/Browser | No | Low | Check home page when notifications are unavailable | Suitable message or empty state appears | To be tested |
| 2 | My Course | Course status | Manual | Invalid course details | Laptop/Browser | No | Low | Open invalid course details | Invalid course is rejected or an error appears | To be tested |
| 3 | Enrollment | Enrollment slot | Manual | Invalid slot | Laptop/Browser | No | Low | Select an invalid slot | Invalid slot is rejected | To be tested |
| 4 | Attendance | Attendance report | Manual | Unavailable data | Laptop/Browser | No | Low | Open unavailable attendance data | Suitable message appears | To be tested |
| 5 | Assignment | Assignment details | Manual | Invalid assignment | Laptop/Browser | No | Low | Open an invalid assignment | Error message appears | To be tested |
| 6 | Examination | Examination details | Manual | Expired link | Laptop/Browser | No | Low | Open an expired examination link | Access is prevented or an error appears | To be tested |
| 7 | Financial Record | Payment history | Manual | Invalid payment details | Laptop/Browser | No | Medium | Enter invalid payment details | Invalid payment is rejected | To be tested |
| 8 | Disciplinary Record | Record access | Manual | Unauthorized account | Laptop/Browser | No | Medium | Attempt unauthorized access | Access is denied | To be tested |
| 9 | Raise Infra Issue | Issue submission | Manual | Empty details | Laptop/Browser | No | Low | Submit an empty issue form | Required fields are indicated | To be tested |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| 10 | My Profile | Profile details | Manual | Invalid profile data | Laptop/Browser | No | Low | Save invalid profile information | Validation message appears | To be tested |

**Result:** Negative test scenarios and expected outputs for the ARMS Portal were documented. Results are marked 'To be tested' because actual execution results were not supplied.

**Code output:**




```
package kbexp;

public class ARMSNegativeTest {

    public static void main(String[] args) {

        String[] items = {
            "Home",
            "My Course",
            "Enrollment",
            "Attendance",
            "Assignment",
            "Examination",
            "Financial Record",
            "Disciplinary Record",
            "Raise Infra Issue",
            "My Profile"
        };

        String[] testCases = {
            "Check missing holiday information",
            "Check invalid course details",
```

**4. To write a case study to identify & analyze the requirements of e-Commerce Applications and to write a positive Test Scenarios as well as Test cases for the above**

**application.**

**Code:**

```
package kbexp;
public class FlipkartNegativeTest {
   public static void main(String[] args) {

      String[] items = {
         "Home",
         "Search Button",
         "Shops",
         "Notifications",
         "Account",
         "Cart",
         "Account Settings",
         "My Cart",
         "Search Product",
         "Order Details"
      };

      String[] testCases = {
         "Open without login",
         "Search with empty input",
         "Open unavailable shop",
         "Check notifications without login",
         "Enter invalid account details",
         "Place order with empty cart",
         "Save profile with missing details",
         "Place order without address",
         "Search for unavailable product",
         "Cancel an invalid order"
      };

      String[] expectedResults = {
         "Ask user to log in if required",
         "Display validation or suitable results",
         "Display suitable message",
         "Ask user to log in if required",
         "Display validation message",
         "Prevent checkout with empty cart",
```

```
            "Validate required fields",
            "Ask user to add a valid address",
            "Display no products found",
            "Display suitable error message"
        };

        System.out.println("FLIPKART - NEGATIVE TEST CASES");
        System.out.println("-------------------------------------------------------------------------");

        System.out.printf("%-5s %-20s %-35s %-35s%n",
                "No.", "Item", "Negative Test Case", "Expected Result");

        System.out.println("-------------------------------------------------------------------------");

        for (int i = 0; i < items.length; i++) {
            System.out.printf("%-5d %-20s %-35s %-35s%n",
                    i + 1, items[i], testCases[i], expectedResults[i]);
        }

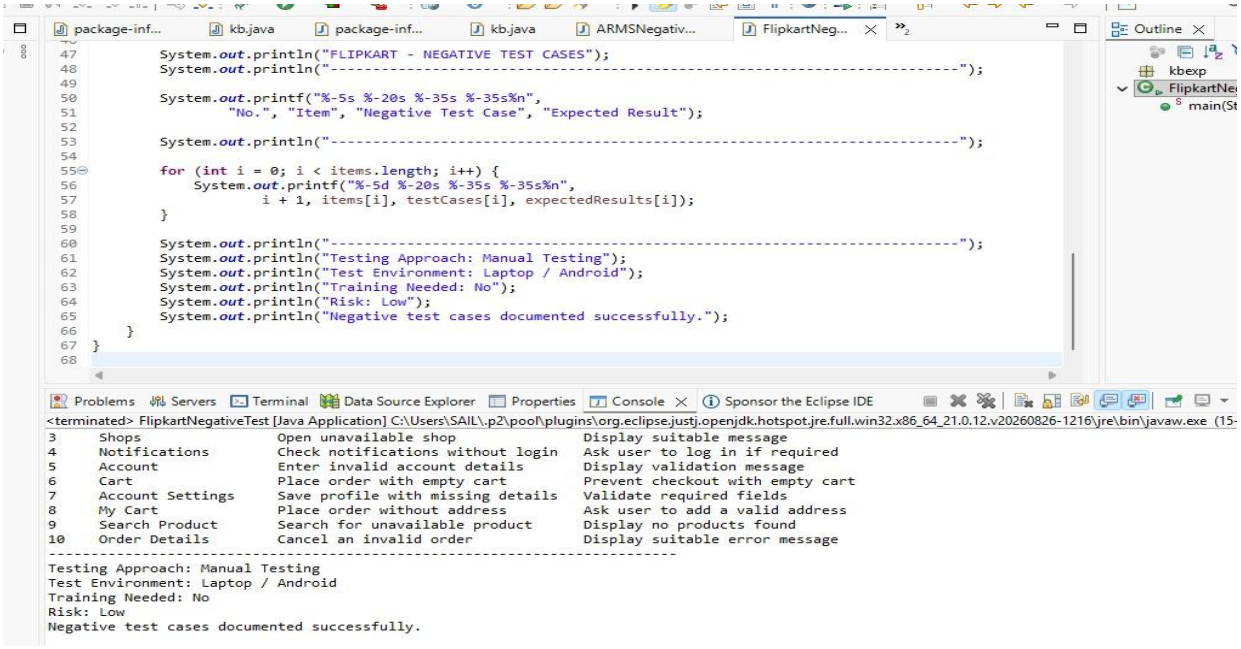
        System.out.println("-------------------------------------------------------------------------");
        System.out.println("Testing Approach: Manual Testing");
        System.out.println("Test Environment: Laptop / Android");
        System.out.println("Training Needed: No");
        System.out.println("Risk: Low");
        System.out.println("Negative test cases documented successfully.");
    }
}
```

**Report:**

| S.No. | Item to be tested | Features | Approach | Test Data | Test Environment | Training Needed | Risk | Test Case / Scenario | Expected Output | Pass/Fail |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Home | Email and password | Manual | Invalid login details | Laptop/Android | No | Low | Open Flipkart and enter incorrect login details | Incorrect email or password message appears | Pass |
| 2 | Search Button | Products, brands and shirts | Manual | Incorrect product | Laptop/Android | No | Low | Search for an incorrect product | Wrong or irrelevant products may appear | Pass |
| 3 | Shops | Mobiles and fashion | Manual | Unavailable mobile | Laptop/Android | No | Low | Search for mobiles in Shops | Searched mobiles do not appear | Pass |
| 4 | Trending Stores | Student and wedding stores | Manual | Store search | Laptop/Android | No | Low | Open a trending store | Another store may open | Pass |
| 5 | Notifications | Order notifications | Manual | Placed order | Laptop/Android | No | Low | Check notifications after placing an order | Order notification does not appear | Pass |
| 6 | Account | Orders, wishlist, coupons | Manual | Account details | Laptop/Android | No | Low | Open Account and Help Centre | Help Centre does not appear | Pass |
| 7 | Account | Logout | Manual | Logged-in account | Laptop/Android | No | Low | Click Logout | Flipkart does not log out | Pass |
| 8 | Account | Edit profile | Manual | Updated profile details | Laptop/Android | No | Low | Edit and save profile | Profile is not updated | Pass |
| 9 | Cart | Place order | Manual | Product in cart | Laptop/Android | No | Medium | Attempt to place an order | Order is not placed | Pass |
| 10 | Shops | Medicines | Manual | Medicine search | Laptop/Android | No | Low | Search for medicines | Medicines do not appear | Pass |

**Result:** Negative test cases for the Flipkart e-commerce application were documented. The Pass labels reproduce the supplied lab-record statuses and are not independently verified.

**Code output:**



```
        System.out.println("FLIPKART - NEGATIVE TEST CASES");
        System.out.println("-------------------------------------------------------------------------------");

        System.out.printf("%-5s %-20s %-35s %-35s%n",
                "No.", "Item", "Negative Test Case", "Expected Result");

        System.out.println("-------------------------------------------------------------------------------");

        for (int i = 0; i < items.length; i++) {
            System.out.printf("%-5d %-20s %-35s %-35s%n",
                    i + 1, items[i], testCases[i], expectedResults[i]);
        }

        System.out.println("-------------------------------------------------------------------------------");
        System.out.println("Testing Approach: Manual Testing");
        System.out.println("Test Environment: Laptop / Android");
        System.out.println("Training Needed: No");
        System.out.println("Risk: Low");
        System.out.println("Negative test cases documented successfully.");
    }
}
```

```
<terminated> FlipkartNegativeTest [Java Application] C:\Users\SAIL\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.12.v20260826-1216\jre\bin\javaw.exe (15-
3     Shops               Open unavailable shop              Display suitable message
4     Notifications       Check notifications without login  Ask user to log in if required
5     Account             Enter invalid account details      Display validation message
6     Cart                Place order with empty cart        Prevent checkout with empty cart
7     Account Settings    Save profile with missing details  Validate required fields
8     My Cart             Place order without address        Ask user to add a valid address
9     Search Product      Search for unavailable product     Display no products found
10    Order Details       Cancel an invalid order            Display suitable error message
-------------------------------------------------------------------------
Testing Approach: Manual Testing
Test Environment: Laptop / Android
Training Needed: No
Risk: Low
Negative test cases documented successfully.
```

**5. To write a case study to identify & analyze the requirements of e-Commerce Applications and to write a negative Test Scenarios as well as Tesi cases for the above**

**application.**

**Code:**

```
package kbexp;
public class FlipkartNegativeTest5 {
   public static void main(String[] args) {

      String[] items = {
         "Home",
         "Search Button",
         "Shops",
         "Trending Stores",
         "Notifications",
         "Account",
         "Logout",
         "Edit Profile",
         "Cart",
         "Medicines"
      };

      String[] testCases = {
         "Enter incorrect email or password",
         "Search for an incorrect product",
         "Search for unavailable mobiles",
         "Open a trending student store",
         "Check notifications for a placed order",
         "Open orders, wishlist and coupons",
         "Click the logout option",
         "Try updating the profile",
         "Try placing an order",
         "Search for medicines"
      };

      String[] expectedOutput = {
         "Incorrect email or password message",
         "Wrong products may appear",
         "Searched mobiles do not appear",
         "Another store may open",
         "Order notification does not appear",
         "Help Centre does not appear",
```

```
            "User is not logged out",
            "Profile is not updated",
            "Order is not placed",
            "Medicines do not appear"
        };

        String[] status = {
            "PASS", "PASS", "PASS", "PASS", "PASS",
            "PASS", "PASS", "PASS", "PASS", "PASS"
        };

        System.out.println("FLIPKART - NEGATIVE TEST CASES");

System.out.println("==========================================================
================");

        for (int i = 0; i < items.length; i++) {
            System.out.println("\nTest Case No: " + (i + 1));
            System.out.println("Item: " + items[i]);
            System.out.println("Test Scenario: " + testCases[i]);
            System.out.println("Expected Output: " + expectedOutput[i]);
            System.out.println("Status in Lab Manual: " + status[i]);
        }

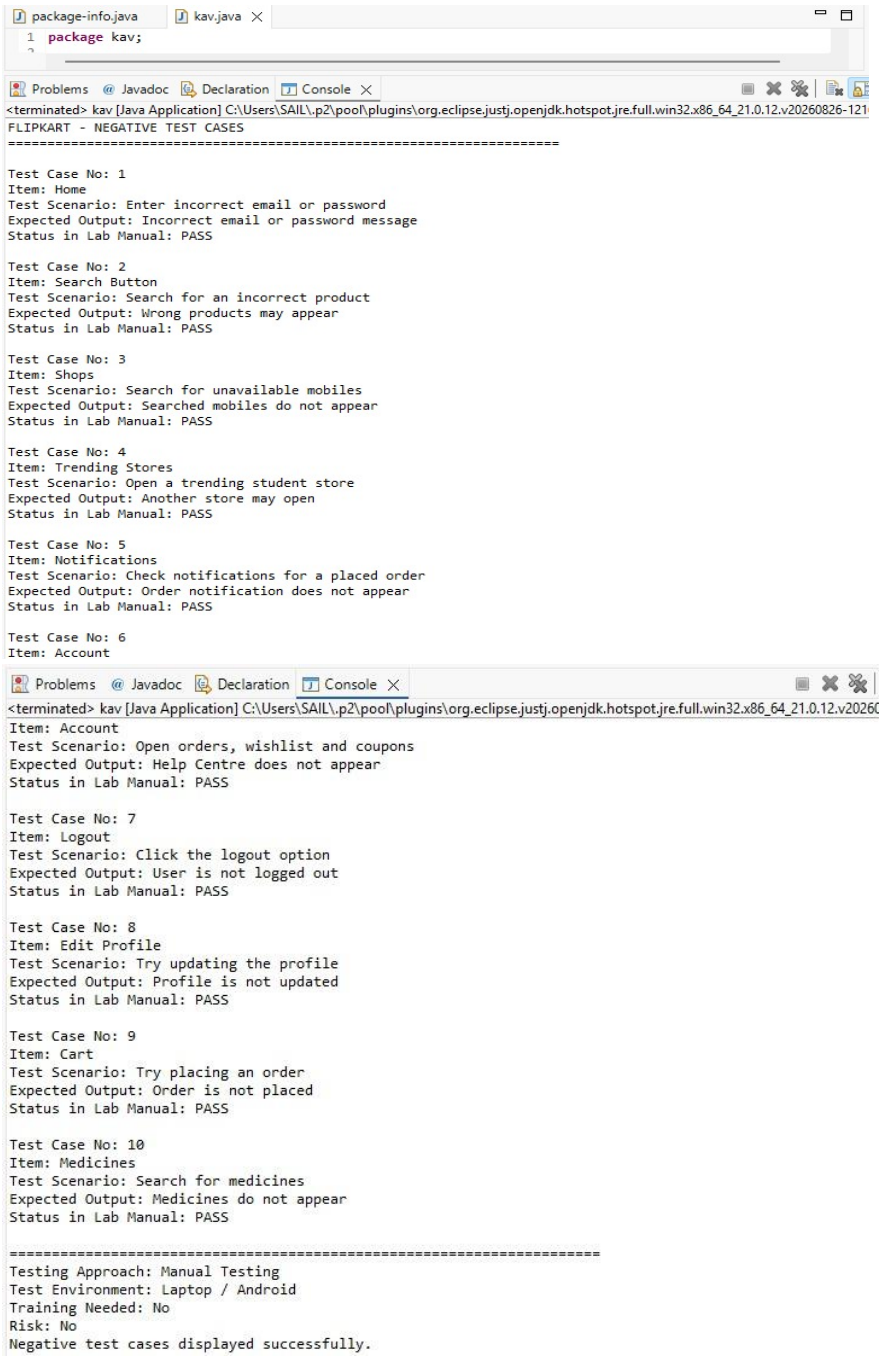

System.out.println("\n==========================================================
================");
        System.out.println("Testing Approach: Manual Testing");
        System.out.println("Test Environment: Laptop / Android");
        System.out.println("Training Needed: No");
        System.out.println("Risk: No");
        System.out.println("Negative test cases displayed successfully.");
    }
}
```

**Report:**

| S. No. | Item to be tested | Features | Approach | Test Data | Test Environment | Training Needed | Risk | Test Case / Scenario | Expected Output | Pass/Fail |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Home | Search, shops, notifications, account | Manual | Invalid email/password | Laptop/Android | No | Low | Open Flipkart and enter incorrect email and password | Incorrect email or password message appears | Pass |
| 2 | Search Button | Products, brands and shirts | Manual | Incorrect product | Laptop/Android | No | Low | Search for an incorrect product | Wrong products may appear | Pass |
| 3 | Shops | Mobiles, fashion, electronics | Manual | Mobile search | Laptop/Android | No | Low | Open Shops and search for mobiles | Searched mobiles do not appear | Pass |
| 4 | Trending Stores | Student and wedding stores | Manual | Store search | Laptop/Android | No | Low | Open Shops and select a trending store | Another store may open | Pass |
| 5 | Notifications | Order list | Manual | Placed order | Laptop/Android | No | Low | Check notifications for a placed order | Order notification does not appear | Pass |
| 6 | Account | Orders, wishlist, coupons | Manual | Account access | Laptop/Android | No | Low | Open Account and check Help Centre | Help Centre does not appear | Pass |
| 7 | Account | Logout | Manual | Logged-in account | Laptop/Android | No | Low | Open Account and click Logout | Flipkart does not log out | Pass |
| 8 | Account | Edit profile | Manual | Updated profile data | Laptop/Android | No | Low | Edit and save profile details | Profile is not updated | Pass |
| 9 | Cart | Place order | Manual | Product in cart | Laptop/Android | No | Low | Attempt to place an order | Order is not placed | Pass |
| 10 | Shops | Medicines | Manual | Medicine search | Laptop/Android | No | Low | Open Shops and search for medicines | Medicines do not appear | Pass |

**Result:** Negative test cases for the Flipkart e-commerce application were documented. The Pass labels reproduce the supplied lab-record statuses and are not independently verified.

**Code output:**


package-info.java
kav.java
1 package kav;
Problems
Javadoc
Declaration
Console
<terminated> kav [Java Application] C:\Users\SAIL\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.12.v20260826-121
FLIPKART - NEGATIVE TEST CASES
Test Case No: 1
Item: Home
Test Scenario: Enter incorrect email or password
Expected Output: Incorrect email or password message
Status in Lab Manual: PASS
Test Case No: 2
Item: Search Button
Test Scenario: Search for an incorrect product
Expected Output: Wrong products may appear
Status in Lab Manual: PASS
Test Case No: 3
Item: Shops
Test Scenario: Search for unavailable mobiles
Expected Output: Searched mobiles do not appear
Status in Lab Manual: PASS
Test Case No: 4
Item: Trending Stores
Test Scenario: Open a trending student store
Expected Output: Another store may open
Status in Lab Manual: PASS
Test Case No: 5
Item: Notifications
Test Scenario: Check notifications for a placed order
Expected Output: Order notification does not appear
Status in Lab Manual: PASS
Test Case No: 6
Item: Account
Problems
Javadoc
Declaration
Console
<terminated> kav [Java Application] C:\Users\SAIL\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.12.v20260
Item: Account
Test Scenario: Open orders, wishlist and coupons
Expected Output: Help Centre does not appear
Status in Lab Manual: PASS
Test Case No: 7
Item: Logout
Test Scenario: Click the logout option
Expected Output: User is not logged out
Status in Lab Manual: PASS
Test Case No: 8
Item: Edit Profile
Test Scenario: Try updating the profile
Expected Output: Profile is not updated
Status in Lab Manual: PASS
Test Case No: 9
Item: Cart
Test Scenario: Try placing an order
Expected Output: Order is not placed
Status in Lab Manual: PASS
Test Case No: 10
Item: Medicines
Test Scenario: Search for medicines
Expected Output: Medicines do not appear
Status in Lab Manual: PASS
Testing Approach: Manual Testing
Test Environment: Laptop / Android
Training Needed: No
Risk: No
Negative test cases displayed successfully.